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The Algorithmic Magistrate: Automated Judicial Orders in Traffic and Municipal Offences and the Constitutional Delegation of Judicial Power

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ABSTRACT:

The proliferation of automated traffic enforcement systems across Indian jurisdictions, particularly the e-challan mechanism operational under the Motor Vehicles Act, 1988, and the Information Technology Act, 2000, has precipitated a constitutional crisis that remains conspicuously under-litigated. This paper undertakes a rigorous examination of the automated adjudicatory apparatus currently deployed in the State of Chhattisgarh—specifically the Integrated Road Accident Database (iRAD) and the e-challan portal managed by the Chhattisgarh Police—to interrogate whether the issuance of automated notices bearing the imprimatur of judicial authority constitutes an impermissible delegation of judicial power under Article 50 and the broader separation of powers doctrine enshrined within the basic structure of the Constitution of India. Through doctrinal analysis of constitutional jurisprudence spanning In Re: Delhi Laws Act (1951) to Madras Bar Association v. Union of India (2021), juxtaposed against a structural examination of the operational architecture of Chhattisgarh's automated traffic enforcement ecosystem, this paper identifies a profound doctrinal lacuna: the legislative authorization for automated notice generation under Section 208 of the Motor Vehicles Act, when operationalized through algorithmic rule-application devoid of human magisterial cognition, effects a de facto transfer of adjudicatory functions from the constitutional court of the Magistrate to the executive apparatus of the Traffic Police. The paper advances the argument that while automation of ministerial functions attendant to judicial proceedings is constitutionally permissible, the algorithmic determination of guilt—even in its inchoate form of a composite notice-cum-order under Section 208—constitutes an unconstitutional erosion of judicial power. The paper concludes by proposing a statutory remediation framework that preserves technological efficiency while mandating a constitutionally compliant human-in-the-loop adjudication mechanism.

Keywords: *Algorithmic adjudication, judicial delegation, e-challan, separation of powers, Motor Vehicles Act 1988, constitutional law, Chhattisgarh, automated legal reasoning.*

1. Introduction

1.1 The Technological Transformation of Summary Adjudication

The architecture of summary criminal adjudication in India stands at a precipice of fundamental transformation. What was once the exclusive province of the Magistrate exercising judicial discretion under the procedural safeguards of the Code of Criminal Procedure, 1973, has been incrementally supplanted by algorithmic systems that issue, process, and effectively adjudicate traffic violations at a scale and velocity that renders meaningful judicial oversight structurally impossible. This modification has occurred not through explicit legislative deliberation on the constitutional implications of automated adjudication, but through the accretion of executive notifications, technical circulars, and software configurations that have, *sub silentio*, reconstituted the very nature of judicial power in the domain of regulatory offences.

The e-challan system, inaugurated under the aegis of the Ministry of Road Transport and Highways and operationalised through state-specific implementations, represents perhaps the most consequential—and least examined—shift in the exercise of judicial power since the establishment of the Supreme Court in 1950. In the State of Chhattisgarh, the Chhattisgarh Traffic Police e-Challan Portal, integrated with the national iRAD database and the VAHAN vehicle registry, processes tens of thousands of alleged violations monthly through an algorithmic pipeline that commences with Automated Number Plate Recognition (ANPR) camera capture and culminates in the generation of a notice that carries, in practical effect if not in formal legal designation, the force of a judicial determination.

1.2 The Constitutional Problem Stated

The constitutional infirmity at the heart of this arrangement may be stated with precision: Article 50 of the Constitution of India mandates that the State shall take steps to separate the judiciary from the executive in the public services of the State. This directive principle, read in harmony with Article 14 (equality before law), Article 21 (right to life and liberty, which encompasses procedural due process), and the basic structure doctrine as propounded in *Kesavananda Bharati v. State of Kerala* (1973) AIR 1973 SC 1461, erects a constitutional firewall between the executive power to investigate and prosecute and the judicial power to adjudicate and punish. The algorithmic issuance of a notice that simultaneously alleges a violation, imposes a monetary penalty, and threatens coercive state action in the event of non-compliance—without the interposition of a judicial mind—raises the foundational question: has the executive, through the instrumentality of code, arrogated unto itself a quantum of judicial power that the Constitution vests exclusively in courts?

This Research paper highlights that question in the affirmative, but with analytical nuance that distinguishes between constitutionally permissible ministerial automation and constitutionally impermissible adjudicatory automation.

1.3 Research Questions

The inquiry revolves around four interconnected research questions:

RQ1: What is the existing precise legal character of an e-challan generated under the Motor Vehicles Act, 1988, read with the Chhattisgarh Motor Vehicles Rules, 1994—is it just a show-cause notice, or composite notice-cum-order, or an adjudicatory Scrutiny?

RQ2: Does the statutory scheme under Section 208 of the Motor Vehicles Act, 1988, as operationalised through the algorithmic infrastructure in Chhattisgarh, effect a delegation of judicial power that transgresses the constitutional limitations established by the Supreme Court's separation of powers jurisprudence?

RQ3: What are the procedural due process deficits inherent in automated enforcement systems that generate legal challenges without human perception of the specific facts of each case?

RQ4: What legislative and technological interventions are necessary to reconcile the administrative efficiency of automated traffic enforcement with the constitutional imperative of judicial adjudication?

1.4 Significance and Scope

This study's importance is not limited to the field of traffic regulation alone. It is a useful case study for the larger trend toward algorithmic governance, i.e., using algorithmic rules to replace human discretion in areas such as determining eligibility for welfare benefits, predicting criminal behaviour, etc. The same constitutional issues that arise from the automated adjudication of traffic violations in Chhattisgarh will also apply to how algorithmic decision-making is implemented throughout the administrative state. How these issues are resolved regarding traffic violations will create guidelines for how the Constitution will apply to AI in Indian public law for many years to come.

The purpose of this research article is to define the boundaries of what will and will not be included within the scope of this paper. Automated adjudication will not be examined in relation to civil matters, and the use of AI in judicial research or case management will be excluded from the materials that will be studied and analysed because they involve separate and distinct questions relating to constitutional law. The focus of this research paper will be restricted exclusively to the automated issuance of citations and penalties for traffic and municipal violations under the current regulatory framework in the state of Chhattisgarh, as a jurisdictional capstone for doctrinal review.

2. Methodology

2.1 Research Design

This paper researches a dual-method research design comprehending doctrinal legal analysis with technical-architectural examination. The doctrinal component involves systematic analysis of constitutional provisions, statutory frameworks, and judicial precedents pertaining to the delegation of judicial power, the separation of powers doctrine, and the procedural requirements for valid adjudication. The technical component involves examination of the operational architecture of the Chhattisgarh e-challan system as disclosed through publicly available government orders, tender documents, system specifications, and the functional characteristics of the public-facing portal.

2.2 Doctrinal Analysis Framework

The doctrinal analysis proceeds along three axes:

Constitutional Axis: Examination of Article 50, Article 14, Article 21, and the basic structure doctrine through the prism of Supreme Court decisions in *In Re: Delhi Laws Act* (1951) AIR 1951 SC 332, *Indira Nehru Gandhi v. Raj Narain* (1975) AIR 1975 SC 2299, *State of Tamil Nadu v. State of Kerala* (2014) 12 SCC 696, and *Madras Bar Association v. Union of India* (2021) 7 SCC 369.

Statutory Axis: Close reading of Sections 207, 208, and 210 of the Motor Vehicles Act, 1988; Rule 11 of the Chhattisgarh Motor Vehicles Rules, 1994; and relevant provisions of the Information Technology Act, 2000 concerning the legal recognition of electronic records.

Procedural Axis: Analysis of the Code of Criminal Procedure, 1973 provisions governing summary trials (Sections 260-265) and the compounding of offences (Section 320), with particular attention to the requirements of judicial cognition and the taking of cognisance under Section 190.

2.3 Technical-Architectural Examination

The technical examination relies exclusively on publicly available and verifiable information sources:

Primary Sources: Chhattisgarh Police notifications establishing the e-challan system; tender documents for the supply, installation, and maintenance of ANPR cameras and associated software (available through the Chhattisgarh e-Procurement Portal); the functional interface and workflow of the Chhattisgarh Traffic Police e-Challan Portal (https://echallan.cgpolice.gov.in); and the integration documentation for the national iRAD system published by the Ministry of Road Transport and Highways.

Secondary Sources: Reports of the Comptroller and Auditor General of India on the implementation of intelligent traffic management systems; parliamentary committee reports on the Motor Vehicles (Amendment) Act, 2019; and academic literature on algorithmic governance and automated legal reasoning.

2.4 Limitations

This study is constrained by three significant limitations. First, the proprietary nature of the software algorithms employed in the ANPR and e-challan generation significantly means that the precise decision logic—the rules by which a camera capture is transformed into a legal notice—is not publicly accessible. The analysis, therefore, proceeds by necessary inference from system outputs and disclosed functional specifications. Second, the absence of published judicial decisions specifically addressing the constitutional validity of automated e-challan adjudication in Chhattisgarh means that the doctrinal analysis operates in a zone of legal uncertainty, extrapolating from general principles to specific application. Third, empirical data on the volume, disposition, and challenge rate of e-challans in Chhattisgarh are not systematically published, limiting the capacity to assess the practical impact of the constitutional concerns identified.

3. The Constitutional Scheme: Judicial Power and Its Impermissible Delegation

3.1 The Contours of Judicial Power Under the Indian Constitution

Although the Constitution of India does not specifically define the expression "judicial power," its fundamental features have been defined by a continuous chain of judicial interpretation over the past seventy years. In Delhi Laws Act 1959 Bench decision, Kania C.J. said that the authority to determine rights and obligations arising from an examination of the facts and law is "essentially judicial in nature" and cannot be delegated by the legislature to the executive, except in respect of collateral or incidental matters involving the delegation of powers. This position was limited by the decision in Hari Shankar Bagla v. Same; based on the same reasoning, the executive cannot be given "essential judicial functions."

For purposes of analysing the essential functions of the judiciary, the core judicial function consists of three irreducible components:

1. **The taking of cognisance:** This involves applying a judicial mind to the allegations that a crime has occurred and establishing whether there are sufficient legal grounds for proceeding.

2. **The reception and evaluation of evidence:** This involves considering the evidence that has been submitted by both prosecution and defence, and determining its probative value based upon applicable legal standards;

3. **The determination of guilt or innocence:** This involves determining whether the facts as established by the evidence warrant the imposition of a sanction as contemplated by law.

The critical analysis difference, which is often overlooked in the discussions concerning automated adjudication, relates to the distinction between purely ministerial functions (or actions) that merely mechanically carry out the predetermined legal consequences, and true adjudicatory (or judicial) functions involving the application of discretionary and judgmental authority or contested issues.

3.2 The Jurisprudence of Tribunalization and Its Limits

Since the adoption of the Constitution, an entirely new branch of law that concerns how executive authorities establish administrative tribunals with authority comparable to the judiciary. This new field arose because there are now many administrative tribunals across India with judicial authority. The Supreme Court, in S.P. Sampath Kumar v. Union of India (1987) 1 SCC 124, decided that the Administrative Tribunals Act, 1985, is valid. In that ruling, the Court stated that the judicial powers exercised by administrative tribunals created under the Act are equal to those exercised by High Courts when they hear civil cases. In the same case, the Court outlined the three minimum constitutional criteria necessary for administrative tribunals to exist under the Constitution: (1) all members of an administrative tribunal must have judicial qualifications and experience; (2) all proceedings conducted by an administrative tribunal must have a judicial nature; and (3) all decisions rendered by an administrative tribunal may be reviewed by a High Court pursuant to Article 226 of the Constitution.

Following the founding principles of jurisprudence, a court is a constitutional floor that cannot go below. There has been recent research examining whether an algorithmic system could serve in an adjudicating capacity. One of the key elements of this inquiry is to define what an "adjudicating body" is. I submit that one must have judicial-type persons with the ability to use judicial judgment; one must follow due process, and there must be adjudications that have the potential for effective court review. Because of this, there is an inherent conflict with algorithmic adjudication -- as an algorithm does not have the attributes of a judge. It cannot listen, weigh competing perspectives, be compassionate or discretionary, or act outside its programmed rules to take into account exceptional circumstances.

3.3 The Distinction Between Legislative, Judicial, and Ministerial Functions

The fundamental justification for distinguishing between allowable automation and disallowed delegation is based on the classification system of three types of government tasks that was established in the case of Jayantilal Amritlal Shodhan v. F.N.Rana, (1964), AIR 1964 SC 648. The Supreme Court made the following observations regarding this case:

"The Legislature may assign authority over some aspects of the policy to the Executive, provided it is not an essential legislative function." Similarly, the Judicial System may not give away its Jurisdiction, unless otherwise provided by Law."

Executive agencies providing Ministerial Services perform a wide variety of ministerial duties, as they act in the process of the judiciary. For example, many of them perform functions necessary to carry on judicial actions, e.g. service of process, preparation of court records, execution of warrants and the collection of fines; these types of duties do not represent the exercise of Jurisdiction, but represent tools used to ensure the performance of judicial activities.

The constitutionality of the e-challan system is based on the question of whether or not an algorithm generates a penalty notice to support prior/succeeding adjudication decisions, or if it is the actual adjudication in and of itself. The determination will determine if all automated enforcement actions were constitutional.

4. The Chhattisgarh E-Challan Paradigm: Architecture and Operation

4.1 Statutory Foundations

Legal authority for automated traffic enforcement in Chhattisgarh rests upon a specific, layered statutory framework. Motor Vehicles Act 1988, updated by the 2019 Amendment, builds the backbone for these systems. Policing agencies rely on Section 207, which grants officers the power to seize vehicles when they suspect traffic infractions. Traffic laws move fast. It is what it is.

Section 208 governs the summary disposal of cases, shifting how courts handle routine movement violations. Truth be told, this structure remains the heart of the enforcement process. Courts taking cognisance of listed offences can issue a summons detailing specific fine amounts. Accused drivers may choose to plead guilty by mail instead of appearing in person before a magistrate. Ignoring red lights or texting while driving feeds this expedited protocol. Judges process these violations quickly to clear heavy backlogs. Actually, the system works.

Reducing mountainous paperwork ensures offenders face immediate consequences for their reckless behaviour. It is what it is, frankly, just business. Detailed schedules define exactly which violations qualify for this simplified approach, leaving little room for ambiguity during ticketing. Everything happens according to the pre-established legislative lists. Documentation matters here. Automated cameras record the evidence, the system generates the summons, and the driver pays the fine under clear statutory guidance. Efficiency is the goal. Drivers comply or pay accordingly. That is the reality.

Section 210 provides the critical procedural mechanism:

"**Section 210 Courts to send intimation about the conviction.**— Every Court by which any person holding a driving licence is convicted of an offence under this Act... shall send intimation to the licensing authority..."

Read together, these provisions authorise a summary procedure wherein the Court—the Magistrate—may, upon taking cognisance, issue a summons that informs the accused of the option to plead guilty by remitting the specified fine without personal appearance. The statutory scheme, on its face, contemplates judicial cognition as the initiating event in the enforcement cascade.

4.2 The Operational Reality: From Camera Capture to Notice Generation

The operational architecture deployed in Chhattisgarh diverges significantly from the statutory model. The system, as disclosed through tender documentation and portal functionality, operates as follows:

Stage One: Automated Detection. ANPR cameras installed at identified junctions in Raipur, Bilaspur, Durg, and other urban centres capture images of vehicles alleged to violate traffic regulations. The camera method makes use of optical character recognition to extract the vehicle registration number from the captured image. This extraction occurs algorithmically, without human verification of image quality, plate legibility, or the accuracy of the alleged violation.

Stage Two: Database Integration. The extracted registration number is queried against the VAHAN national vehicle registry and the SARATHI driving licence database to retrieve the registered owner's name and address. This retrieval is automated and proceeds without human verification of the accuracy of the database entry or the currency of the ownership information.

Stage Three: Notice Generation. The system generates a composite document titled "Traffic Violation Notice" or "e-Challan." This document, transmitted to the registered owner via SMS to the mobile number linked in the VAHAN database and simultaneously posted to the registered address, contains: (a) an allegation that a specified violation occurred at a specified time and place; (b) a demand for payment of a specified fine amount; (c) a statement that the matter will be referred to Court upon non-payment; and (d) a threat that the driving licence may be suspended and the vehicle registration may be cancelled.

Stage Four: Adjudication by Default. The system treats non-payment within the stipulated period as an election to contest, whereupon the matter is ostensibly referred to the jurisdictional Magistrate. However, the practical reality—documented in reports of the Comptroller and Auditor General concerning similar systems in other states—is that the volume of generated challans far exceeds the adjudicatory capacity of the Magistracy. The referral is frequently ministerial: the Court receives a bulk electronic file and issues process through the same automated pipeline.

4.3 The Judicial Cognition Deficit

The critical constitutional defect in all this is the lack of a judicial mind determining when you become legally culpable. The summons is the legal document that generates legal responsibility and threatens punishment by the government; the summons must be issued by "the Court taking cognisance" under the statutory authority of section 208. Under section 190 of the Code of Criminal Procedure, 1973, taking cognisance of a complaint is a judicial act, whereby the Judicial Officer takes notice of the evidence presented to them and determines whether there is sufficient evidence to proceed with the case.

In the (Chhattisgarh) e-challan system (sued for failure to ensure), there is no application of a judicial mind as to any violation (alleged) prior to the generation of the legal consequence. For example, the system creates a legal notice stating whether the vehicle captured on ANPR exceeded the speed limit, passed the stop line, or broke some other regulation based solely on an algorithm that determined these actions by using specific data stored at particular locations. Although the state is threatening punishment based on the notice to the property owner, it is generating the notice by a computer system created and run by the executive police.

This is not simply a procedural error, but rather a shuffling of judicial authority between the constitutionally established courts and executive agencies of government. The magistrate has now been placed into a significantly diminished position as they will be called upon to act only when the individual subject to the computerised judgment contests it. However, for most individuals who receive an electronic challan (i.e., the individuals who pay the amount due without contesting) there is no involvement by the courts during any aspect of the enforcement process, from finding someone guilty through to imposing a penalty.

5. The Delegation Doctrine Under Scrutiny: Automated Adjudication as Impermissible Transfer of Judicial Power

5.1 The *Madras Bar Association* Framework and Its Application

In Madras Bar Association v. Union of India (2021) 7 SCC 369, the Supreme Court held invalid and unconstitutional the Tribunal Reforms (Rationalisation and Conditions of Service) Ordinance, 2021. The court also provided additional details concerning the limitations on the delegation of judicial power by way of specific principles articulated by the court in the Constitution Bench ruling.

There are two principles set forth by the Supreme Court that must be taken into consideration in determining the outcome of this inquiry, as they directly relate to it.

1. Judicial power belongs solely and exclusively to the constitutional courts. The executive cannot delegate its judicial power to other authorities, whether it is a tribunal or other authority, unless such delegation is specifically authorised by the Constitution itself.

2. The delegation of adjudicatory power from the executive to tribunals, such as the one created by the Ordinance, is not permitted unless the tribunals were created so as to provide for judicial independence through security of tenure and freedom from executive control.

The primary attributes of a court (rendering final determinations, requiring the attendance and production of witnesses/documents, and sanctioning for contempt of court) cannot be conferred on entities without the institutional characteristics of independent judicial administration.

5.2 The Chhattisgarh System as *De Facto* Adjudication

The characterisation of the e-challan notice as a mere "show-cause notice" requiring the recipient to respond—a characterisation that the State of Chhattisgarh would presumably advance in defence of the system—fails to account for the functional reality of the enforcement mechanism. The following factors establish that the e-challan system in Chhattisgarh operates as a *de facto* adjudicatory apparatus:

Chhattisgarh system notices behave like multifaceted legal documents. Honestly, these records do more than just flag a traffic breach. One single file presents a firm claim of wrongdoing while simultaneously locking in the exact penalty amount due. Such precise documentation leaves drivers with fewer options for dialogues

5.3 The Impermissible Delegation Argument Formulated

Legal critics argue that the Chhattisgarh e-challan architecture suffers from a specific constitutional flaw. India vests judicial authority strictly within established courts. Executing the power to judge guilt or apply criminal penalties belongs solely to the judiciary, yet this system delegates those duties directly to the executive branch. Algorithms effectively hunt for targets and calculate fines automatically.

Actually, the machine acts as judge and jury. That is the reality. By functioning this way, the process breaches Article 50, Article 14, and Article 21, ignoring the basic structure doctrine protected by our supreme laws. Claiming a recipient maintains the right to contest does not fix the underlying injury. Why? Because the current system treats an automated guess as an enforceable truth before any judge reviews the evidence.

Shifting the burden of proof onto the average citizen creates a massive imbalance. State power requires prior judicial backing to function under the rule of law, rather than relying on an unverified algorithmic charge that a victim must laboriously disprove. Drivers face automatic penalties for traffic violations as states bypass

oversight. Presuming guilt defaults to punishing citizens, forcing everyone to scramble for relief instead of getting a fair trial. Actually, officials prioritise speed over rights daily. Keep moving. That reflects our grim, cold reality.

Actually, quick digital processing cannot replace the necessity of fair, independent legal scrutiny in criminal matters. Everything feels broken when machines decide who pays today. That is the hard reality for everyone.

6. The Unexplored Frontier: Algorithmic Adjudication and the Future of Judicial Power

6.1 The Phenomenology of Algorithmic Judgment

The constitutional analysis has been done so far within the conventional doctrinal paradigms. However, the advent of algorithmic adjudication raises deeper questions about the nature of judicial judgment that legal theory has scarcely begun to address. What does it mean to "adjudicate" when the act of judgment is performed by a computational system executing a fixed rule-set? Is there a qualitative difference that the Constitution must recognise and protect, between application of law by a human judge and application of law by an algorithm?

This is not an argument for leniency in traffic enforcement. It is an argument about the nature of law. Law, in its highest conception, is not a system of inflexible commands backed by sanctions. It is a normative order that depends on its legitimacy on the perception that its application is mediated by human judgment capable of recognising the particular in light of the universal. When we replace the human magistrate with the algorithmic magistrate, we do not simply make enforcement more efficient. We change the character of the legal order itself.

6.2 The Procedural Due Process Deficits

The procedural due process implications of automated traffic adjudication in Chhattisgarh extend beyond the delegation doctrine to encompass fundamental requirements of natural justice. Article 21 of the Constitution guarantees that no person shall be deprived of life or personal liberty except according to procedure established by law. The Supreme Court has repeatedly held that **"procedure established by law"** must be **"right and just and fair and not arbitrary, fanciful or oppressive"** (*Maneka Gandhi v. Union of India* (1978) 1 SCC 248).

The Chhattisgarh e-challan system manifests multiple procedural deficits that render its procedure arbitrary and oppressive:

Deficit One: The Inability to Cross-Examine the Accuser. The recipient of an e-challan is confronted with an accusation made by a machine. There is no human accuser to cross-examine, no opportunity to test the

reliability of the detection apparatus through adversarial questioning. The recipient cannot ask: When was this camera last calibrated? What is its margin of error? Was the image reviewed by a human operator before the notice was generated? These questions go to the heart of the reliability of the accusation, yet the system provides no mechanism for their resolution.

Deficit Two: The Presumption of Instrument Accuracy. The system depends on an irrebuttable presumption that the ANPR camera, the optical character recognition software and the database integration pipeline are all functioning perfectly. Statistical evidence from jurisdictions that have mature automated enforcement systems (the United Kingdom, Australia and the United States) shows this presumption is unfounded. Errors in camera calibration, misreading of plates, inaccuracies in databases, and software glitches create a non-trivial rate of false notices. Chhattisgarh is a relatively new system, and the protocols for quality control are not public knowledge, so the error rate is unknown but unlikely to be zero.

Deficit Three: The Burden of Technological Sophistication. Contesting an automated notice requires a degree of technological sophistication that is not uniformly distributed across the population. The recipient must navigate the e-challan portal, locate the contested notice, upload supporting documentation, and articulate a defence in terms that the system can process. For citizens lacking digital literacy—a demographic that includes the elderly, the economically marginalised, and residents of rural areas—the contestation mechanism is effectively inaccessible. The system thus discriminates, in practical effect, against those least equipped to navigate its demands.

6.3 The Equality Clause Implications

Article 14 of the Constitution guarantees equality before the law and equal protection of the law. The Supreme Court has laid down this guarantee to prohibit not only intentional discrimination but also laws and practices that, while facially neutral, produce discriminatory effects (*Navtej Singh Johar v. Union of India* (2018) 10 SCC 1).

The Chhattisgarh e-challan system raises two distinct Article 14 concerns:

First, the system creates a two-tiered adjudicatory mechanism based on economic capacity. Those who can afford to pay the fine without contestation—even if they believe the allegation to be erroneous—subsidise a system that imposes disproportionate burdens on those who cannot. The economically secure motorist pays ₹500 and forgets the matter; the economically precarious motorist faces the Hobson's choice of paying a fine she cannot afford or undertaking a contestation process she cannot navigate.

Second, the system's reliance on the VAHAN database for owner identification creates a presumption of liability that falls with particular severity on certain categories of vehicle owners. A vehicle owned by a

corporation, a government department, or a fleet operator may be driven by any number of individuals. The registered owner's liability for violations committed by unknown drivers is a form of vicarious liability that the Motor Vehicles Act does not impose and that Article 14 cannot sustain absent a rational nexus to a legitimate state purpose.

7. Remedial Pathways: Reconciling Efficiency with Constitutionalism

7.1 The Human-in-the-Loop Imperative

The constitutional infirmities identified in this paper are not incurable. The efficiency gains of automated traffic enforcement are substantial and legitimate state interests. The challenge is to design a system that preserves these efficiency gains while complying with the constitutional imperatives of judicial adjudication and procedural due process.

The essential remedial principle is the human-in-the-loop requirement. At a minimum, no legal consequence—no fine, no penalty, no threat of coercive action—should attach to an automated detection until a human judicial officer has applied her mind to the material and determined that there exists a *prima facie* case.

This requirement can be satisfied through a redesigned workflow:

Step One: Automated Detection and Capture. The ANPR system captures images of alleged violations and extracts relevant data. This data is transmitted to a central repository but does not, at this stage, generate any notice or communication to the registered owner.

Step Two: Magistrate Review. A designated Magistrate—perhaps a Special Metropolitan Magistrate appointed for this purpose—reviews a batch of captured violations. The review need not be exhaustive; it need only verify that: (a) the image is of sufficient quality to identify the vehicle; (b) the alleged violation is discernible from the image; and (c) the registration number is legible. The Magistrate then electronically endorses the issuance of process.

Step Three: Notice Generation. Only upon such judicial endorsement does the system generate and transmit the Section 208 notice. The notice should bear the electronic signature of the Magistrate, indicating that cognisance has been taken.

Step Four: Contested Proceedings. Where the recipient contests the notice, the matter proceeds to a summary trial before the Magistrate, with the recipient allowed cross-examining the prosecution's witnesses, including the technician responsible for camera calibration and maintenance.

7.2 Statutory Amendment Requirements

The proposed remedial architecture requires statutory amendment. Section 208 of the Motor Vehicles Act, 1988, should be amended to clarify that the "Court taking cognisance" must be a judicial officer and that the algorithmic detection of a violation does not, by itself, constitute the taking of cognisance. The amendment should further provide that no notice under Section 208 shall issue except upon the order of a Magistrate.

Additionally, the Information Technology Act, 2000, should be amended to address the specific evidentiary challenges posed by machine-generated evidence in criminal proceedings. Section 65B, which governs the admissibility of electronic records, should be supplemented by provisions requiring the prosecution to produce certification of: (a) the calibration status of the detection apparatus; (b) the error rate of the optical character recognition software; and (c) the integrity of the data pipeline from capture to notice generation.

7.3 Institutional Safeguards

Beyond statutory amendment, institutional safeguards are essential to ensure the integrity of automated enforcement. These include:

Independent Technical Audit. The Chhattisgarh e-challan system should be subject to an annual independent technical audit by an agency unconnected to the Traffic Police. The audit should assess camera calibration, software accuracy, data integrity, and the rate of erroneous notices. The audit report should be published and made available to recipients seeking to contest notices.

Publicly Accessible Calibration Records. The Traffic Police should maintain and publish calibration records for each ANPR camera, including the date of last calibration, the calibration standard applied, and the margin of error certified.

Notice of Rights. Every e-challan should prominently inform the recipient of: (a) the right to contest the notice without payment; (b) the procedure for contestation; (c) the availability of legal aid for economically disadvantaged recipients; and (d) the fact that payment constitutes an admission of guilt only if made with full knowledge of the right to contest.

8. Conclusion

An algorithm-based approach to summary adjudication in the traffic enforcement system in Chhattisgarh is a turning point for the interpretation of the Constitution. There are compelling reasons for automation in terms of efficiency. There is a large backlog of traffic cases at the Magistracy; there are limits to the resources available; and it is in the public interest to have safe roads. However, these factors do not provide adequate justification for violating the fundamental tenets of the Constitution.

The e-challan system currently used in Chhattisgarh creates an unconstitutional delegation of judicial authority. The executive branch has been given the ability to determine guilt and impose punishment through algorithms, with only a theoretical right to appeal that is, in reality, inaccessible to the majority of people. This system assumes that individuals are guilty until proven innocent, replaces a system of judicial review with an executive action, and changes the traditional practice of judicial review.

Basically, what I am trying to say is that while some may view the remedial path to justice as backwards technologically or too much for the courts to handle, it is not either of those things at all when we consider a human-in-the-loop architecture that retains the benefits of automated detection through judicial determination of consequence. It is possible to strike a balance between efficiency and constitutionalism through this type of architecture, even though there will be costs (time spent by judges, money spent by administrators or extra time taken to enforce) associated with the establishment of a constitutional system of government.

This case also poses a broader question about what type of legal system India aspires to be; specifically, should there be one where the state can punish its citizens based only upon the use of algorithm-generated penalties or where there is a human intermediary that serves as an adjudicator of how the state penalises its citizens? Both of these are legal systems that can be constitutional; however, it will require that we consider the vision for India's legal system and ultimately decide what type of legal order we want to create as part of establishing a constitutional government. Therefore, choosing between these two systems is more than just an administrative or technical issue; it strikes deeply at our constitutional foundation.

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